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AI-Supported Teaching Material Development in Higher Education

Exploring Teachers’ Practices and Perceptions of Environmental Sustainability

Degree Project in Informatics IK7004, 30 credits

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Abstract

Artificial intelligence (AI) is becoming more common in higher education, including in the development of teaching materials. While previous research has examined the pedagogical uses of AI and its environmental implications, less attention has been given to how teachers use AI in developing teaching materials and how they understand its environmental consequences. This study explores these issues through semi-structured interviews with 12 higher education teachers from different disciplinary and geographical contexts. The findings show that teachers used AI for lesson planning, assessment design, generating ideas, classroom activities, and organizing teaching materials. AI was valued for saving time and supporting idea generation, but teachers continued to rely on their professional judgment to review, adapt, and evaluate AI-generated content. Their AI literacy developed mainly through practical experience and self-directed learning, with limited formal training. Environmental sustainability was less familiar to teachers, with most reporting limited awareness of AI's environmental impacts. Some teachers had nevertheless begun to avoid unnecessary interactions, reuse existing materials, and use AI when it served a clear teaching purpose. Teachers also expressed a need for greater knowledge and institutional support to understand these issues. The study shows that teachers' professional judgment remains central to AI-supported teaching material development, while environmental sustainability is still developing as a consideration in the use of AI.

Keywords: Artificial intelligence, Environmental sustainability, Higher education teachers, Teaching material development, Teaching practices.

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1. Introduction

Digital transformation has become an integral part of higher education, and the emergence of Artificial Intelligence (AI) technologies has influenced how universities approach teaching, learning, and administrative processes (Lytras et al., 2024). These technological developments have changed conventional teaching and learning approaches. In particular, AI has gained significant attention for its potential to support a wide range of educational activities. Therefore, the incorporation of AI in higher education is becoming more prevalent in teaching and learning practices (Griche & Bennis, 2026). AI can support different aspects of teaching and learning, including personalized learning and adaptive feedback (Zhai et al., 2021; Zhai, 2023). However, its growing role in supporting teachers is particularly relevant to this study, as AI is used to prepare lesson planning, assessment design, and the development of learning materials (Ramkissoon, 2024).

The use and value of AI in education depend not only on its technical capabilities but also on how teachers understand, evaluate, and apply it within their teaching practices. Recent research indicates that teachers' perceptions, professional knowledge, and experiences influence how AI tools are integrated into higher education settings (Georgiou et al., 2026). This highlights the important role teachers play in deciding how AI is used in their teaching practice.

While using AI provides opportunities to support teaching practices, its growing use also raises concerns about environmental sustainability. Digital technologies require physical infrastructure, energy consumption, and material resources, which contribute to environmental impacts such as carbon emissions and electronic waste (Hilty & Aebischer, 2015). AI systems also require computational resources for their development and operation. As the use of AI expands, it raises concerns about energy consumption and resource demands (Aczel et al., 2026). Therefore, discussions about AI integration in higher education need to consider educational applications alongside the environmental implications associated with its use.

Daniel et al. (2026) emphasize the importance of considering the environmental implications of AI alongside its potential benefits in higher education. This is particularly relevant as AI becomes increasingly integrated into teaching and other university activities. Teachers' perceptions and sense of responsibility can also influence how AI is approached and used in higher education (McGrath et al., 2023). Since teachers decide when, how, and for which teaching tasks AI is used, their decisions may also shape whether environmental sustainability is considered in its use. Decisions about the frequency and purpose of AI use may have environmental implications. Despite growing interest in AI and environmental sustainability, less is known about how higher education teachers consider environmental sustainability when using AI to develop teaching materials. Understanding teachers' perceptions and experiences can therefore provide insight into whether and how environmental considerations are incorporated into decisions about the use of AI in teaching practice.

1.1 Purpose of the Study

This study investigates how higher education teachers use AI to develop teaching materials and how they perceive environmental sustainability in relation to AI use. It focuses on teachers' experiences and views on using AI to develop teaching materials, with particular attention to how they understand and consider the environmental implications of AI use.

Research Question

This study is guided by the following research questions

RQ1:
How do higher education teachers use AI to develop teaching materials?

RQ2: How do higher education teachers perceive the environmental sustainability implications of using AI to develop teaching materials?

1.2 Significance of the study

This study brings together two related areas: AI-supported teaching practices and environmental sustainability. Existing research has examined the pedagogical potential of AI, including its use in teaching-related tasks, personalized learning, feedback, and the development of educational resources (Zhai et al., 2021; Zhai, 2023; Ramkissoon, 2024). Recent research has also examined the potential of AI in sustainability education, including its use to support learning experiences and engagement with complex sustainability challenges (Leal Filho et al., 2025). However, while the educational applications of AI have received increasing attention, less attention has been given to the environmental implications associated with its use in teaching, particularly in the development of teaching materials. By exploring teachers' experiences and perceptions, this study contributes to discussions about the responsible integration of AI by considering both its pedagogical value and environmental implications.

The findings may be useful for higher education institutions, educators, and policymakers. They may inform professional development, institutional AI policies, and discussions about the responsible and sustainable use of AI in education. The study also provides insight into how teachers make decisions about using AI in teaching and whether environmental considerations form part of these decisions.

1.3 Delimitations

This study is limited to teachers' self-reported experiences and practices regarding the use of AI for developing teaching materials. The study does not evaluate the technical performance of AI systems, measure energy consumption, or calculate carbon emissions. Instead, it focuses on teachers' experiences, perspectives, and reflections on the use of AI in education.

The study focuses on higher education teachers and other stakeholders, such as students, administrators, policymakers, and technology developers, who are not included. These

delimitations show that the study focuses on teachers' experiences and perspectives regarding AI use and environmental sustainability in higher education.

2. Related Work

This section reviews research relevant to the two main aspects of the study: teachers' use of AI in teaching and in material development, and the environmental sustainability implications of its use. It first considers how teachers use AI in teaching-related activities and the role of AI in developing teaching materials. It then examines research on the environmental impacts of AI and how environmental sustainability has been discussed in higher education. The section also considers responsible AI, with particular attention to teachers' professional judgment and the role of institutional support. These areas are brought together to identify what is currently known about teachers' use of AI and environmental sustainability and to establish the research gap addressed by this study.

2.1 AI Use in Teaching and Material Design

Higher education teachers use AI tools to support various teaching-related activities, such as lesson preparation, instructional planning, and the development of teaching materials (Zhai, 2023). Teachers use a range of generative AI, such as ChatGPT, Gemini, DALL·E, Microsoft Bing Image Creator, Synthesia, and Canva, depending on their instructional needs and learning objectives (Kılıçkaya & Kic-Drgas, 2025). Previous research indicates that teachers use AI for various tasks, including generating teaching content, designing learning activities, preparing lesson plans, developing classroom resources, and supporting assessment-related activities (Kılıçkaya & Kic-Drgas, 2025; Oh, 2025; Saxena & Wei, 2025).

Studies across different educational contexts illustrate how teachers incorporate AI into their teaching practices. For instance, Saxena and Wei (2025) found that pre-service teachers used AI to create stories, worksheets, and personalized learning materials. Similarly, Kılıçkaya and Kic-Drgas (2025) reported that language teachers used AI to prepare lesson plans and classroom exercises, while Oh (2025) showed that teachers used AI for lesson design, classroom activities, and the creation of visual learning materials. Recent studies in higher education further indicate that university teachers use AI to prepare teaching materials, generate content, develop rubrics and prompts, support feedback practices, and assist with administrative tasks (Baggia et al., 2026; Zaim et al., 2024). At the same time, the increasing use of AI by students is creating new considerations for teachers' teaching and assessment practices. Han et al. (2025), in their review of research on detecting GenAI-generated writing, found that educators have difficulty distinguishing AI-generated work from student-produced work, particularly when AI-generated content is subsequently edited or modified. Their findings suggest that the growing use of AI is not only influencing how teachers prepare learning materials but also shaping how they approach assessment and evaluate student work.

These studies show that AI is becoming part of a range of teaching-related activities, from preparing instructional materials to responding to changes in student learning and assessment practices. However, existing research has largely concentrated on the pedagogical uses and potential benefits of AI, while less attention has been given to the environmental considerations associated with its use. Although some studies acknowledge sustainability-related concerns, the environmental implications of AI use in educational contexts remain relatively underexplored (Oh, 2025). This leaves a limited understanding of whether and how higher education teachers consider environmental sustainability when using AI in their teaching practices, including when developing teaching materials.

2.2 Environmental Impact of AI

The environmental impact of AI has become an important topic in recent research. Oliveira et al. (2026) argue that environmental consequences can arise throughout the development, deployment, and use of AI systems. Relatedly, according to Aczel et al. (2026), AI systems require substantial computing resources, contributing to energy consumption, carbon emissions, water use, and land use. Their report also highlights that data centers supporting AI systems consume large amounts of water for cooling, with some located in regions already experiencing drought or water scarcity. Similarly, Kar et al. (2022) note that training and operating AI models require considerable computing power, which contributes to higher energy consumption and greenhouse gas emissions. These environmental impacts can also be understood in terms of a carbon footprint, which provides a way to estimate the greenhouse gas emissions associated with particular activities and sources, including the use of digital technologies in higher education (Valls-Val & Bovea, 2021).

These studies denote that AI systems have environmental implications that extend beyond their technical and educational applications. As AI tools are used in higher education, it is relevant to consider whether educators are aware of the environmental implications of these technologies. However, existing research provides limited insight into higher education teachers' perceptions of AI's environmental impact and whether such considerations influence their decisions when using AI to develop teaching materials. Therefore, further research is needed to understand how educators perceive and consider the environmental implications associated with AI use in higher education.

2.3 Environmental Sustainability in AI-Supported Higher Education

Environmental sustainability has become an important consideration in discussions of AI-supported teaching and learning in higher education (Hassan & Ahmad, 2025). As AI is incorporated into educational practice, attention has also been given to how its use aligns with broader sustainability goals. Similarly, Hamadi and El-Den (2024) argue that sustainability should be integrated into digital learning and institutional decision-making, suggesting that decisions about adopting digital technologies should consider environmental responsibility alongside educational objectives. In addition, Valls-Val and Bovea (2021) argue that higher education institutions should assess the environmental impacts of their activities to support

sustainability planning and more environmentally informed decision-making. These studies indicate that environmental sustainability is becoming part of discussions about the adoption and use of digital technologies in higher education

Although AI provides opportunities to support teaching and learning, its increasing use has also raised questions about environmental sustainability. In this context, Leal Filho et al. (2025) acknowledge the educational applications of AI while emphasizing that universities should also consider the environmental implications of these technologies as it becomes more widely adopted. Likewise, Daniel et al. (2026) argue that empirical evidence on the long-term environmental implications of AI adoption in higher education remains limited. Existing research highlights the need to consider environmental sustainability alongside the educational use of AI in higher education.

However, existing research has mainly examined these issues from institutional, technological, or policy perspectives. Comparatively little attention has been given to the perspectives of higher education teachers, even though they make decisions about how AI is used in teaching activities, including the development of teaching materials. Understanding how teachers perceive environmental sustainability in relation to AI use can deepen understanding of how environmental considerations are taken into account in everyday teaching practice in higher education.

2.4 Responsible AI and Environmental Sustainability

Responsible AI use has mainly been discussed in relation to ethical principles such as fairness, transparency, and accountability. However, recent discussions have also included environmental sustainability, as concerns about the environmental effects of AI use have increased. Johnson (2025) argues that responsible AI should consider the environmental consequences of AI and encourage more sustainable practices. Extending this discussion, Aczel et al. (2026) emphasize that every use of AI requires natural resources and that its environmental impact depends on how systems are designed, deployed, and used. They highlight the importance of responsible choices throughout the AI lifecycle, including the efficient use of computational resources and consideration of the environmental consequences of AI-related decisions at both individual and institutional levels.

In higher education, responsible and sustainable AI use is also shaped by institutional support. Georgiou et al. (2026) show that teachers rely on institutional guidance, training, and clear policies when integrating generative AI into their teaching. Their TPACK-based study indicates that teachers need appropriate frameworks and professional development to make responsible decisions about AI use and address related ethical and practical issues.

At the same time, responsible AI requires human judgment in teaching practice. Mota et al. (2026) highlight that although teachers' roles are changing with the introduction of AI, professional judgment remains important. Their review shows that educators evaluate AI outputs, decide how AI is used, and ensure that AI-supported teaching aligns with pedagogical,

ethical, and professional expectations. This shows that responsible AI involves not only institutional policies but also teachers' decisions in their teaching practice.

Although responsible AI has been widely discussed in relation to ethics, governance, and institutional approaches, less attention has been given to how teachers consider environmental sustainability when using AI in teaching. Understanding teachers' perceptions can provide insight into how environmental considerations are included in responsible AI use in higher education.

2.5 Synthesis and Research Gap

The reviewed literature shows the use of AI in higher education to support teaching-related activities, including lesson preparation, instructional planning, content creation, and the development of teaching materials (Zhai, 2023; Kılıçkaya & Kic-Drgas, 2025; Oh, 2025; Baggia et al., 2026). Previous studies demonstrate that AI can assist teachers with generating learning resources, designing activities, preparing teaching materials, and supporting assessment-related tasks. At the same time, research has highlighted the need for teachers to evaluate AI-generated content and exercise professional judgment when using such outputs (Han et al., 2025). However, existing research has mainly focused on the pedagogical applications, uses, and potential benefits of AI, with limited attention given to the environmental considerations associated with its use in teaching practice.

Research on the environmental impact of AI shows that AI systems require substantial computational resources and are associated with energy consumption, carbon emissions, water use, and other environmental pressures (Kar et al., 2022; Aczel et al., 2026). These concerns have extended discussions of AI beyond educational outcomes to include questions of environmental sustainability and responsible use. In higher education, existing studies highlight the importance of considering environmental sustainability when adopting digital technologies and developing approaches to responsible AI use (Hamadi & El-Den, 2024; Leal Filho et al., 2025; Johnson, 2025). However, these discussions have not sufficiently addressed how teachers understand and apply environmental considerations in their use of AI.

Much of the existing literature examines environmental sustainability and responsible AI from institutional, technological, or governance perspectives. Studies have paid less attention to how higher education teachers perceive environmental sustainability and whether these perceptions influence their decisions when using AI in teaching activities. This is important because teachers make practical decisions about when and how AI is used, how generated content is evaluated and adapted, and whether AI-generated materials are considered appropriate for particular teaching contexts. Although teachers have an important role in shaping the use of AI in higher education, there is limited understanding of how they perceive and consider its environmental implications in their teaching practices.

Despite growing research on AI-supported teaching, environmental sustainability, and responsible AI, these areas have largely been examined separately. There is a limited

understanding of how teachers connect these issues when using AI to develop teaching materials. Therefore, this study explores higher education teachers' perceptions of environmental sustainability in the development of AI-supported teaching materials. By examining teachers' experiences and perspectives, the study contributes to a better understanding of how environmental considerations are reflected in the everyday and responsible use of AI in higher education. Table 1 presents an overview of the literature reviewed in this section, highlighting the main focus areas, key findings, and research gaps that inform the current study.

Table 1

Summary of the Reviewed Literature, Key Findings, and Research Gap

Table 1

Summary of the Reviewed Literature, Key Findings, and Research Gap

Focus Area	Key Studies	Main Findings and Research Gap
AI Use in Teaching and Material Design	Zhai (2023); Kılıçkaya & Kic-Drgas (2025); Oh (2025); Saxena & Wei (2025); Baggia et al. (2026); Han et al. (2025)	Research shows that AI is used to support a range of teaching-related activities, including lesson preparation, instructional planning, content creation, learning activities, teaching material development, and assessment-related tasks. Research also points to challenges in evaluating AI-generated content and highlights the continuing role of human judgment when working with such content. However, existing research has largely focused on pedagogical applications and potential benefits, with less attention to the environmental considerations associated with teachers' use of AI in everyday teaching practice.
Environmental Impact of AI	Kar et al. (2022); Valls-Val & Bovea (2021); Oliveira et al. (2026); Aczel et al. (2026)	Research identifies environmental pressures associated with AI systems, including energy consumption, carbon emissions, water use, and land use. These studies establish the environmental implications of AI beyond its technological and educational applications. However, limited research has examined how higher education teachers perceive these environmental implications or whether such considerations influence their use of AI for teaching and material development.

Environmental Sustainability in AI-Supported Higher Education	Hamadi & El-Den (2024); Hassan & Ahmad (2025); Valls-Val & Bovea (2021); Leal Filho et al. (2025); Daniel et al. (2026)	Existing research emphasises the importance of considering environmental sustainability when adopting and integrating AI and other digital technologies in higher education. The literature also points to the need to consider environmental implications alongside educational objectives. However, much of this discussion is situated at institutional, strategic, or broader educational levels, with comparatively limited attention to teachers' perspectives and everyday practices.
Responsible AI and Environmental Sustainability	Johnson (2025); Aczel et al. (2026); Georgiou et al. (2026); Mota et al. (2026)	Recent discussions of responsible AI increasingly extend beyond ethical concerns to include environmental sustainability. The literature also highlights the importance of institutional guidance, professional development, and teachers' professional judgment in responsible AI use. However, limited attention has been given to how higher education teachers themselves consider environmental sustainability when making decisions about AI use in teaching practice.

3. Theoretical Framework

This study examines how higher education teachers use AI to develop teaching materials and how they perceive the environmental sustainability implications of this use. These two aspects require different theoretical perspectives. While one framework is needed to understand how teachers integrate technology into their teaching practice, another is needed to explore how they recognize and reflect on the environmental consequences of technology use. Therefore, this study draws on two complementary frameworks: the Technological Pedagogical Content Knowledge (TPACK) framework (Mishra & Koehler, 2006) and Clark's (2024) concept of Critical Climate Awareness.

3.1 Technological Pedagogical Content Knowledge (TPACK)

The TPACK framework (Mishra & Koehler, 2006) was selected because this study focuses on how teachers integrate AI into their teaching practices. The framework explains how teachers combine technological, pedagogical, and subject knowledge when making decisions about technology use. This perspective is relevant because AI can support the creation of lesson plans, presentations, activities, and other teaching resources; however, teachers remain responsible for evaluating whether AI-generated materials are accurate, appropriate for students, and aligned with learning objectives.

The use of AI in teaching material development involves more than technical ability. It requires teachers to apply their pedagogical understanding and subject expertise when creating, adapting, and reviewing AI-supported materials. In this study, TPACK is not used to measure teachers' levels of technological, pedagogical, or content knowledge. Instead, it provides an interpretive lens for understanding how teachers describe their experiences integrating AI into the development of teaching materials and how they combine AI capabilities with their professional knowledge.

3.2 Critical Climate Awareness

Although TPACK helps explain teachers' use of AI in teaching practice, it does not address the environmental consequences associated with technology use. Since this study also examines environmental sustainability considerations, Clark's (2024) concept of Critical Climate Awareness is used as a complementary perspective.

Clark (2024) argues that environmental awareness involves recognizing the relationship between everyday actions and their wider environmental consequences. This concept is relevant to the present study because teachers may use AI as a digital teaching tool without necessarily considering the resources and environmental impacts associated with its operation. Although AI applications appear intangible, their development and use depend on computational resources, energy consumption, and digital infrastructure.

In this study, Critical Climate Awareness is used to explore how teachers understand the environmental implications of AI use and whether these considerations influence their decisions when developing teaching materials. The framework extends the analysis beyond the practical use of AI by examining teachers' awareness and reflection regarding the sustainability consequences of their technology-related choices.

3.3 Integration of the Frameworks

The two frameworks address different but connected aspects of this study. TPACK provides a basis for understanding how teachers integrate AI into the development of teaching materials, while Critical Climate Awareness helps explain how teachers perceive and reflect on the environmental implications of AI use. Together, these frameworks allow the study to examine teachers' experiences from both pedagogical and sustainability perspectives.

4. Research Design

This study adopted a qualitative exploratory research design to examine how higher education teachers use AI to develop teaching materials and how they perceive the environmental sustainability implications of this use. A qualitative approach was appropriate because the study focuses on teachers' experiences, interpretations, and reflections within their professional contexts (Merriam & Tisdell, 2017). The aim was not to measure the extent of AI adoption but to understand how teachers describe and make sense of their experiences. An exploratory

design was selected because the relationship between AI-supported teaching material development and environmental sustainability in higher education remains underexplored. When knowledge about a phenomenon is still developing, an exploratory approach can provide a deeper understanding and help identify areas for further investigation (Dekkers et al., 2022).

4.1 Data Collection

Semi-structured interviews were selected because they provide a balance between consistency across participants and flexibility to explore individual experiences in greater depth (Kallio et al., 2016). This flexibility makes semi-structured interviews particularly suitable for exploratory qualitative research, where the focus is on gaining an in-depth understanding of participants' experiences and perspectives through open-ended and flexible discussion. (Merriam & Tisdell, 2017; Dekkers et al., 2022). In the present study, this approach enabled participants to describe their AI-supported teaching practices and their perceptions of environmental sustainability in detail. An interview guide was used to ensure that similar topics were covered across all interviews while allowing participants to elaborate on issues they considered important (Kallio et al., 2016). Based on this approach, semi-structured interviews were conducted with 12 participants between March and April 2026. The interviews were conducted online via Google Meet because participants were located in different geographic regions, and this format provided flexibility in scheduling. Each interview lasted approximately 45–60 minutes and was audio-recorded with participants' informed consent. All interviews were transcribed verbatim, and the transcripts were manually reviewed for accuracy before the data analysis. Table 2 provides an overview of the participants, including their teaching area, years of experience, geographical region, and interview duration.

The interview guide was developed with reference to the study's theoretical framework. Questions concerning AI use reflected the pedagogical focus of the TPACK framework (Mishra & Koehler, 2006), while questions related to environmental sustainability were informed by Clark's (2024) concept of critical climate awareness. These perspectives helped ensure consistency across the study's topics, theoretical framework, and data collection. The complete interview guide is provided in Appendix A.

4.2 Participants and Sampling

Purposive sampling was the primary recruitment strategy, with participants selected for their experience using AI in teaching practice. This approach enables researchers to recruit participants who can provide relevant and meaningful insights into the phenomenon under study (Merriam & Tisdell, 2017; Campbell et al., 2020). Snowball sampling was used as a complementary approach when participants recommended other teachers who met the inclusion criteria (Campbell et al., 2020). The recruitment process began by sending interview invitations to higher education teachers who met the inclusion criteria. The invitation explained the purpose of the study, what participation involved, and that participation was voluntary. Teachers who agreed to participate were interviewed, and several participants recommended colleagues who also met the inclusion criteria. These recommendations formed the basis of

snowball sampling, which helped recruit additional participants with relevant experience (Campbell et al., 2020).

A total of 12 higher education teachers participated in the study. The participants represented different academic disciplines and had varying levels of teaching experience, providing a range of perspectives on the use of AI in higher education. To protect participants' identities, each participant was assigned a code (T1–T12), where "T" denotes teacher. These codes are used throughout the thesis, including Table 2 and the presentation of the findings. An overview of the participants' characteristics is presented in Table 2. The sample size was guided by the purpose of the study and the need to obtain rich, detailed accounts from participants with relevant experience, consistent with qualitative research (Merriam & Tisdell, 2017).

Table 2

Participant Information

Participant ID	**Teaching Area**	**Years of Experience**	**Country**	**Interview Duration (min)**
T1	Special Education & Teacher Education	25	Sweden	58
T2	English Language and Linguistics	7	India	55
T3	Informatics & Digital Design	14	Sweden	60
T4	Education & English Language Teaching	8	India	52
T5	History	20	India	56
T6	Teacher Education & Educational Technology	26	Denmark	60
T7	Physics	11	India	55
T8	German Language & Special Needs Education	11	Sweden	50
T9	English Literature	7	India	48
T10	Commerce	6	India	52
T11	English Literature	12	India	57

T12	Learning Disabilities & Teacher Education	14	Sweden	57

4.3 Data Analysis

The interview data were analysed using Braun and Clarke's (2006) six-phase thematic analysis approach. The six phases involved (1) familiarising oneself with the data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes, and (6) producing the report. Braun and Clarke (2006) describe thematic analysis as a flexible method for identifying, analyzing, and reporting patterns within qualitative data. It was considered appropriate for this study because the aim was to explore higher education teachers' experiences with using AI tools to develop teaching materials and their perceptions of environmental sustainability. Consistent with Merriam and Tisdell (2017), the analysis focused on interpreting participants' experiences to develop an understanding of the phenomenon under investigation.

Before the analysis began, the interview recordings were transcribed using Microsoft 365. The transcripts were then manually reviewed while listening to the audio recordings to correct any transcription errors and ensure their accuracy. The verified transcripts were read several times to familiarise the researcher with the data, and preliminary notes were taken to capture initial ideas and recurring patterns regarding AI use, teaching practices, and environmental sustainability.

The transcripts were then manually coded by identifying meaningful sections of the data relevant to the research question. The analysis combined inductive coding with theory-informed interpretation, where themes were developed from participants' accounts and later considered in relation to TPACK and Critical Climate Awareness. Related codes were grouped into broader categories to develop preliminary themes, which were subsequently reviewed and refined by comparing them with both the coded extracts and the complete dataset. Where necessary, themes were reorganized, combined, or separated to improve their clarity and ensure they accurately represented participants' experiences and addressed the research question. Examples of the coding process are presented in Appendix B.

In the final stages of the analysis, each theme was clearly defined and named according to its central meaning. The final themes were interpreted in relation to the research questions and theoretical framework, with representative quotations used to illustrate the findings. The relationships among the final themes are illustrated in Figure 1, while the coding process and theme development are presented in Appendix B.

4.4 Ethical Considerations

Ethical considerations are essential in research involving human participants, ensuring that participants' rights, dignity, and well-being are protected throughout the research process. This

study adhered to the principles of good research practice outlined by the Swedish Research Council (2025).

Before data collection, participants received an information and consent letter explaining the purpose of the study, what participation involved, how the collected data would be used, and their right to withdraw from the study at any time without consequences. The information and consent letter are included in Appendix C. It was shared before the interviews to allow participants sufficient time to read the information and ask questions before deciding whether to participate. At the beginning of each interview, participants were reminded of the study's purpose and ethical conditions, and explicit consent was obtained before audio recording.

Participants' identities were protected through anonymity and confidentiality. Each participant was assigned an alphanumeric code (e.g., T1–T12), which was used throughout the analysis and reporting instead of names or other identifying information. Audio recordings and interview transcripts were stored on password-protected devices accessible only to the researcher. The handling of personal data followed the principles of the General Data Protection Regulation (GDPR) and the Swedish Research Council's (2025) guidelines. Identifiable information was removed during transcription, the data were used solely for academic purposes, and all identifiable data will be securely deleted following the final approval of the thesis.

The interview guide focused on participants' professional experiences and did not include sensitive or intrusive questions, thereby minimising the risk of psychological, professional, or personal discomfort. Throughout the research process, reflexivity was maintained by critically considering how the researcher's assumptions and interpretations could influence the analysis. Participants' perspectives were represented as accurately and respectfully as possible throughout the reporting of the findings.

5. Findings

This section presents the findings organized into five themes. The first four themes focus on how higher education teachers use AI to develop teaching materials, including its role as a pedagogical support tool, the importance of teacher expertise in evaluating and adapting AI-generated content, the development of AI literacy through experience, and the pedagogical considerations that shape AI-supported material development. The fifth theme examines teachers' perceptions of environmental sustainability and responsible AI use in relation to the development of teaching materials. Together, these themes address the two research questions guiding the study. The findings are organized into the following five themes:

1. AI as a Pedagogical Support Tool
2. Teacher Expertise for Evaluating and Validating AI Outputs
3. Developing AI Literacy Through Practice, Adaptation, and Support

4. Pedagogical Considerations in AI-Supported Teaching Material Development
5. Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development

Figure 1

Thematic Map of Teachers' Practices in AI-Supported Teaching Material Development and Environmental Sustainability

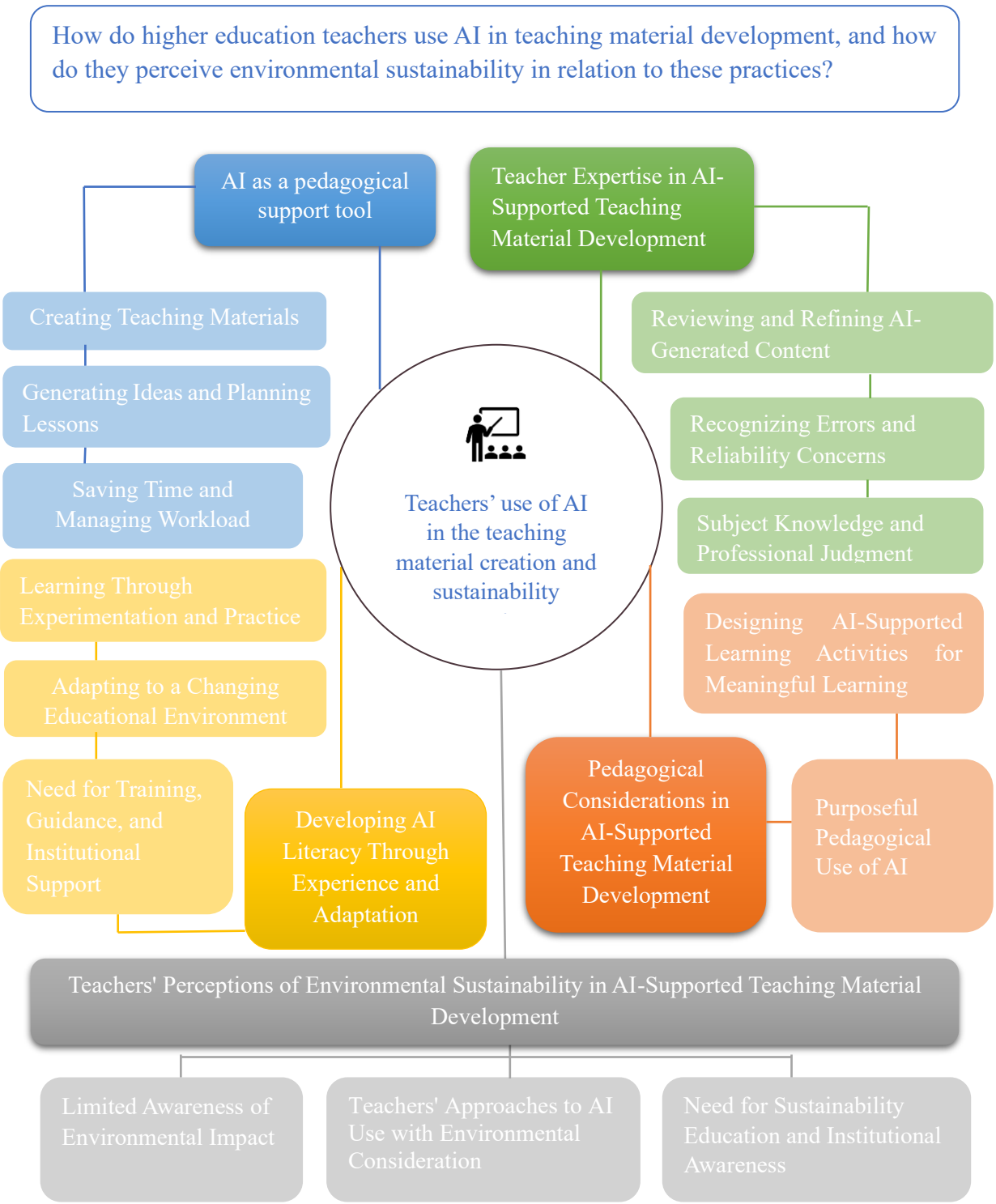


Figure 1 presents a thematic map showing the main themes identified from the data and their relationships to AI-supported teaching material development and environmental sustainability.

5.1 AI as a Pedagogical Support Tool

Teachers use AI as a pedagogical support tool to develop teaching materials and teaching-related activities. During the interviews, teachers reported that they use AI to develop teaching materials, brainstorming ideas for lesson planning, and to reduce workload in teaching-related tasks. Although teachers used AI in various ways across disciplines and teaching contexts. Teachers consistently described it as a useful tool for preparing teaching materials.

5.1.1 Creating Teaching Materials

Teachers reported using AI to prepare various types of teaching materials, including lesson plans, assessment questions, classroom activities, presentations, worksheets, case scenarios, and other instructional resources. Although the applications differed across subjects, teachers adapted AI-generated content according to their teaching objectives and classroom requirements.

Several teachers mentioned how AI simplified the preparation of educational materials. For example, T5 explained that AI helped with the development of assessment questions and made the preparation process more efficient, as "I use AI to assist me in creating the class test questionnaires, and made the preparation process easier and more efficient" (T5).

Teachers also used AI to develop subject-specific teaching materials. A special education teacher noted that AI helped generate realistic case scenarios that students could analyze and discuss during classroom activities, encouraging them to consider how they would respond in realistic situations. According to T1, "I use it to create cases (realistic scenarios) for my teaching." Similarly, T11 reported using AI to prepare lecture materials, creating presentation slides for a literature course. "I recently used AI to create a PowerPoint for the Romantic Age of English literature."

The findings show that teachers used AI in different ways to support the development of teaching materials. Although the specific uses varied across teachers and subjects, they described using AI for a range of teaching-related purposes based on their teaching needs.

5.1.2 Generating Ideas and Planning Lessons

In addition to developing teaching materials, teachers described using AI as a source of ideas during lesson planning. They explained that they turned to AI when they needed new ideas, different perspectives, or alternative ways of developing teaching materials. One teacher said that AI was particularly useful when motivation was low or when it was difficult to come up with new ideas, stating that, "I mostly turn to AI when I'm low on motivation, running out of ideas, and feeling stuck" (T1). T4 expressed a similar view, stating, "When I feel tired, lack of ideas, and when I feel some kind of stuck, I use AI."

Teachers also stated that AI enabled them to consider different perspectives when planning learning activities. As mentioned by T7, "I could explore more from many perspectives to include activities in my class, and it works and makes the class more engaging."

These reflections show that teachers used AI as a brainstorming tool during lesson planning and the development of teaching materials. Participants described using AI to generate ideas, explore different perspectives, and consider alternative approaches when planning their teaching.

5.1.3 Saving Time and Managing Workload

Teachers consistently mentioned saving time as one of the main benefits of using AI for teaching material development. They noted that preparing teaching materials, assessments, and classroom activities required considerable time and effort, particularly alongside other teaching and administrative responsibilities. They reported that AI helped streamline these routine tasks and reduce the time needed to prepare teaching materials.

Several teachers described AI as helping them save time during teaching material preparation. For example, T1 stated, "It saves time, especially when you have a lot to do." Similarly, T9 described AI as reducing workload and improving efficiency, stating, "It really helped me reduce my workload and save time and energy."

Conversely, teachers also recognized that AI did not always reduce workload. When the generated content was inaccurate, incomplete, or unsuitable for a particular teaching context, additional time was needed to review, edit, or rewrite the material. According to T11, "When the content is not what I need, then I have spent more time on editing and recreating it." (T11)

The findings indicate that teachers viewed AI as a practical tool for improving efficiency in teaching material development. However, the time-saving benefits depended on teachers' ability to evaluate, modify, and adapt AI-generated content before using it in their teaching practice.

5.2 Teacher Expertise for evaluating and validating AI outputs

Although teachers viewed AI as a useful tool for developing teaching materials, they emphasized that AI-generated content required careful evaluation before use in teaching. Teachers described reviewing, editing, and adapting AI-generated materials to ensure that the content was accurate, relevant, and suitable for their subject areas and teaching contexts before using it in teaching. Their accounts indicate that AI supported the development process, but teachers remained responsible for judging the quality and appropriateness of the final materials.

5.2.1 Reviewing and Refining AI-Generated Content

Teachers stated that reviewing and modifying AI-generated content is an important part of their process for developing teaching materials. They explained that they checked whether the generated information matched their lesson objectives, students' learning needs, and classroom context before incorporating it into their teaching materials.

Several teachers explained how they evaluated AI-generated materials before use. For instance, T2 described revising prompts and modifying generated content when necessary. As T2 explained, "I always try to read and check the provided materials, and if needed, I recreate and modify my prompt."

Similarly, T5 highlighted the importance of carefully reviewing AI-generated information, stating, "I always check the generated content, and I read it twice very slowly."

These responses show that teachers viewed AI-generated materials as resources that required further review and adaptation. They did not simply transfer AI outputs directly into their teaching but used their own judgment to improve the suitability of the materials.

5.2.2 Recognizing Errors and Reliability Concerns

Teachers also raised concerns about the reliability of AI-generated content. They reported encountering factual inaccuracies, misleading explanations, incorrect references, and other errors when using AI to develop teaching materials.

One teacher recalled noticing a spelling error in the name of a well-known researcher, which raised concerns about the reliability of the entire response. "When I saw that spelled it wrong, I needed to go through the whole text" (T1). T7 also highlighted the importance of using AI cautiously, stating, "We should be very conscious of the content we generate as it can make mistakes."

These experiences influenced how teachers approached AI-generated content. Their experiences with factual inaccuracies, incorrect references, and other mistakes made them more cautious when using AI to develop teaching materials. They described the need to verify information and assess its accuracy before using it in teaching.

5.2.3 Subject Knowledge and Professional Judgment

Teachers emphasized that subject knowledge and professional judgment were important when working with AI-generated content. They explained that their expertise helped them identify inaccuracies, evaluate the relevance of information, and determine whether materials were appropriate for their students and learning objectives. They also highlighted that subject knowledge was necessary to identify mistakes and verify AI-generated content. As T1 explained, "You should have the knowledge within the subject to identify the mistakes."

Teachers also explained that AI-generated content should be reviewed critically because it could contain biased or inaccurate information. They emphasized the importance of evaluating AI outputs before using them in teaching. As T9 stated, "We can't use AI blindly as sometimes it can make inappropriate... so our pedagogical skill and subject knowledge also have a bigger part in the process."

These responses show that teachers relied on their subject knowledge and professional judgment when reviewing AI-generated content. Their subject knowledge and pedagogical judgment guided their evaluation, adaptation, and decision about whether AI-generated content was appropriate for classroom use.

5.3 Developing AI Literacy Through Practice, Adaptation, and Support

Teachers explained that their understanding of AI developed gradually through continued use in their teaching practice. They stated that their learning occurred mainly through experimentation, repeated use, self-directed learning, and discussions with colleagues. Over time, these experiences helped them become more familiar with AI tools and better understand how to apply them in the development of teaching materials. The findings indicate three interconnected aspects of this learning process. First, teachers developed their AI literacy through experimentation and practice. Second, they adapted to the changing role of AI in higher education and recognized the need to keep their knowledge up to date. Third, they highlighted the importance of institutional guidance and professional development to support their continued development of AI-related skills.

5.3.1 Learning Through Experimentation and Practice

Teachers primarily developed their AI skills through hands-on experience, experimentation, and repeated practice. Since opportunities for formal training were limited, many participants learned by exploring different AI tools, testing prompts, and gradually improving their ability to generate useful responses.

Some teachers shared that their early stages of using AI as challenging because they were unfamiliar with the technology and unsure how to interact with it effectively. As stated by T12, "I really needed lots of time… while I was adapting to technology." Teachers also explained that learning how to write effective prompts was an important part of becoming more confident AI users. According to T9, "From the beginning, I was facing difficulty with how to give the prompt." Over time, teachers explained that making mistakes helped them understand how to communicate with AI more effectively. Through repeated experimentation, they learned how to write clearer prompts and obtain more relevant responses. As T2 noted, "I learn from the mistake and try not to make it next time." Similarly, T8 mentioned that the experience improved the quality of interactions with AI, "From mistakes, I learned how to give detailed and refined prompts (T8)."

These experiences show that teachers developed AI literacy through an ongoing process of experimentation and adjustment. Their ability to use AI effectively was shaped by repeated interaction, reflection on experiences, and gradual improvement in their use of AI tools.

5.3.2 Adapting to a Changing Educational Environment

Beyond learning through experience, teachers noted the need to adapt to the growing presence of AI in higher education. As AI became more common in teaching, they felt it was important to understand how it worked and how to use it when developing teaching materials. Several teachers explained that gaining this understanding had become increasingly relevant, not only because AI is now part of educational practice but also because it is becoming more prominent in wider society.

Some teachers argued that AI was a technology that was getting harder to ignore. As T4 pointed out, " I need to learn it, because AI is a growing term, and it is being used in every field…

otherwise I feel I am far away from the current trend." Few teachers explained that their engagement with AI was influenced by professional expectations and the need to stay informed about developments affecting higher education. Their decision to use AI is motivated by the pressure of professional demands, as T3 stated, "I feel like I am forced to use AI because you have to know how it works."

These responses show that teachers' interest in learning AI was influenced not only by personal curiosity but also by changes within their professional environment. Teachers perceived learning about AI as necessary to remain relevant and adapt to evolving expectations in higher education.

5.3.3 Need for Training, Guidance, and Institutional Support

Although teachers developed AI-related skills through personal exploration and experience, they also stressed the need for additional institutional support. Teachers noted during the interviews that they depended on personal experimentation, online resources, and informal discussions with colleagues because structured professional development opportunities were limited.

More teachers expressed a desire to expand their knowledge of AI and its educational applications. As T1 stated, "I need more knowledge… I need to read more about how to use AI." Similarly, T5 emphasized the need for continued learning, as "We need more knowledge on this topic (AI use in education)." Teachers also point out the limited availability of institutional guidance on the use of AI in the development of teaching materials. As T3 expressed, "I don't think the institution has anything as far as I know."

Furthermore, teachers suggested that workshops, seminars, and opportunities to share experiences with colleagues could support their continued learning. As T7 noted, “A teachers' forum would be beneficial, where we can share our experiences of using AI with other teachers. I believe it can help others to learn from that.” T3 also stressed the need for workshops, as “I think workshops and seminars can help us to get more knowledge on this topic (AI use in education).”

These findings indicate that teachers were actively developing their AI literacy, but they also recognized that individual effort alone may not be sufficient. Institutional support, professional development, and opportunities for collaboration were viewed as important factors in helping teachers use AI more confidently and effectively in developing teaching materials.

5.4 Pedagogical Considerations in AI-Supported Teaching Material Development

While the previous themes explored teachers' experiences of adopting and learning to use AI, this theme shows the pedagogical considerations that influenced how AI was used to develop teaching materials. Teachers carefully considered how AI-supported materials could be developed to support student learning. Their decisions extended beyond simply developing teaching materials efficiently to ensuring that AI-generated content aligned with learning

objectives, classroom environment, and meaningful learning experiences. These considerations influenced how teachers designed learning activities, developed teaching materials, and integrated AI into their teaching practice. Their experiences suggest that AI was used to support teaching in ways that reflected their pedagogical goals and professional judgment.

5.4.1 Designing AI-Supported Learning Activities for Meaningful Learning

Although teachers frequently used AI to support the development of teaching materials, they also reflected on how AI influenced the design of learning activities and assessments. Teachers explained that students' increasing use of AI required them to reconsider how learning activities could encourage participation, critical thinking, and meaningful learning. Several teachers expressed concern that excessive reliance on AI might reduce students' engagement with the learning process, leading them to adapt their teaching materials and assessment activities accordingly.

As stated by T9, "Students use AI for their assignments, so it is difficult to determine whether they have used it." Similarly, T12 remarked on the purpose of assessment, "The ultimate aim for the assignments is student learning… but now there is doubt whether learning is really happening."

In response to these concerns, teachers described how they developed teaching materials that encouraged students to engage more actively with the course content. For example, T9 said, "I make questionnaires with higher-order thinking levels to lower-level thinking so students can reflect and engage more with the topic."

T8 pointed out, “Since students also use AI, I need to be more careful when designing teaching materials and learning activities. The tasks should promote meaningful learning and encourage students to engage with the content instead of depending solely on AI-generated responses.”

These remarks suggest that teachers considered more than the efficiency of AI when developing teaching materials. They also reflected on how AI-supported learning activities could promote active learning, encourage reflection, and ensure that students achieve the intended learning outcomes**.**

5.4.2 Purposeful Pedagogical Use of AI

Teachers further stated that effective AI-supported teaching material development required purposeful planning and professional judgment. Instead of merely using AI simply to generate teaching materials, teachers described using AI purposefully, adapting AI-generated materials to align with learning objectives, students' diverse learning needs, and the specific context of their classes. They emphasized that defining learning goals and providing clear prompts helped ensure that AI-generated materials were relevant to their teaching context and supported students' learning. As T12 stated, "I use AI tools with defined goals. I always use them to gain better results, so I am always specific with my needs before using them."

Teachers also highlighted the importance of critical awareness and remaining focused on their teaching objectives instead of following every suggestion generated by AI. One teacher

claimed that “When I use AI, it often suggests additional ideas and suggestions that I didn’t request. Although these suggestions can be helpful, they can also shift my attention away from my intended task, and I spend more time. So, I think we should have clarity upon what we need so use AI with a clear purpose” (T9).

These findings show that teachers used AI purposefully to support teaching material development. Although AI assisted with generating content, teachers decided how that content would be used in their courses. They adapted AI-generated materials to their learning objectives, students' needs, and classroom context, while designing learning activities that supported meaningful student learning. These responses indicate that teachers remained responsible for making the pedagogical decisions that shaped the use of AI in teaching.

5.5 Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development

This theme explores teachers' perceptions of environmental sustainability in relation to their use of AI for developing teaching materials. Most teachers reported having limited knowledge of the environmental impacts associated with AI and expressed a need to learn more about the topic. Despite this limited awareness, many believed that environmental sustainability should be considered when AI is used in educational practice. Although several teachers were uncertain about how AI affects the environment, some described ways they tried to use AI with environmental considerations in mind. Teachers also highlighted the need for more awareness, institutional guidance, and sustainability education to support more informed use of AI.

5.5.1 Limited Awareness of Environmental Impact

Most teachers reported having limited knowledge of the environmental impacts associated with AI. They explained that discussions about AI within their institutions and professional contexts focused mainly on plagiarism, academic integrity, student use, and educational efficiency, while environmental sustainability received much less attention.

Some teachers described uncertainty about the relationship between AI and environmental sustainability. For example, T10 stated, "I was wondering what the connection between the environment and AI is, and we often discuss student use and plagiarism with colleagues." Similarly, T11 acknowledged having some awareness of the issue but a limited understanding of its environmental effects: "I have known about AI and ethical consideration, but I really don’t know how it is connected with the environmental consequences." T9 was similarly uncertain about the environmental costs of AI use, stating, "I know there is probably some environmental impact, but I honestly do not know how much energy or resources are used for a single chat or video." These responses suggest that although teachers recognized that AI may have environmental consequences, they had only a limited understanding of how AI systems consume energy and contribute to those impacts. For many teachers, environmental sustainability has received much less attention than issues such as plagiarism, academic integrity, and educational efficiency.

Teachers also observed that environmental sustainability was rarely discussed within their institutions, professional training, and conversations about AI in teaching. Instead, discussions tended to focus on issues such as plagiarism, ethics, student use, and educational efficiency. These responses indicate that environmental sustainability has not yet become a prominent part of discussions surrounding AI use in higher education.

5.5.2 Environmental Considerations in Teachers' AI Use

Although most teachers reported limited knowledge of AI's environmental impact, some described ways they considered environmental sustainability when using AI to develop teaching materials. They explained that they used AI only when it served a clear teaching purpose and avoided using it for casual or unnecessary tasks. One teacher described this as a personal approach to using AI with environmental consideration. As T8 explained, "I don't use it for fun or unnecessary things. I always use it with a clear purpose. This is what I follow in my use of AI with consideration for environmental impact, but I am not sure how effective it is." (T8). Similarly, T7 explained that their approach to AI use was based on personal understanding instead of a formal strategy, stating, "I don't follow any specific strategy. I usually use it with caution. It is based on my understanding, not any proven strategy. So I think we should use it consciously and carefully. Such critical awareness is important while using AI. We can reduce the time of engagement with AI, otherwise we need to spend lots of time making changes and altering the content." (T7).

These responses suggest that some teachers described using AI in ways they believed reflected environmental considerations, such as using it only when necessary and avoiding unnecessary use. They also indicated that these practices were guided by their own understanding rather than formal guidance or institutional policies.

Some teachers also described practical approaches they followed when using AI. These included reusing previously developed teaching materials, avoiding unnecessary prompts, and limiting the generation of AI-generated images when not required. As T3 explained, "As responsible use with environmental consideration, I don't generate new AI images every time. I reuse existing ones and create new images only when required for the lesson. I think reusing the content can be an effective step for environmental consideration." Similarly, T6 stated, "I try to give clear, simple, and detailed prompts so I don't have to keep changing them. It saves time, and I feel that writing clear prompts can be a useful practice for reducing unnecessary use of AI. This is something I follow with environmental consideration."

These responses indicate that some teachers adopted practical ways of using AI that they believed reflected environmental consideration. These included using AI only when necessary, reusing existing teaching materials, limiting the generation of AI-generated images, and writing clear prompts to reduce repeated interactions with AI. However, teachers also explained that

these approaches were based largely on their own judgment and experience rather than formal knowledge or institutional guidance.

5.5.3 Need for Sustainability Education and Institutional Awareness

Many teachers highlighted the need for more awareness, education, and institutional discussion about AI and environmental sustainability. They explained that the environmental implications of AI were not widely understood or discussed within higher education.

Several teachers emphasized that both educators and students needed more information about how the use of AI affects the environment. "Some people still don't know how it is impacting the environment, so it is important to give more focus and discussions (T8)." Another teacher suggested that AI and its environmental impact should be more directly incorporated into education, as T10 claimed that "We should include AI and its impact awareness topics in the curriculum." Teachers also highlighted the importance of considering environmental sustainability for the future. As T11 pointed out, "It is important to think and act wisely for a sustainable future."

These responses suggest that teachers viewed greater awareness of AI's environmental implications as important for both educators and students. They also highlighted the need for institutional support through professional development, curriculum guidance, and opportunities for discussion. For example, T11 stated, "I feel we need some workshop, training, or guidance from the institution side so we can use AI in a more environmentally sustainable way. As a team, we can make a change." Likewise, T3 emphasized the value of continued discussion, explaining, "We need more discussion and knowledge on this topic within our institutions, which can improve the use of AI and environmental consideration."

Overall, these findings indicate that teachers viewed institutional support as important for strengthening their understanding of the environmental implications of AI. They suggested workshops, professional development, curriculum integration, and opportunities for discussion to increase awareness and support more environmentally informed use of AI in the development of teaching materials.

6. Discussion

The findings demonstrate the important role of teachers in using AI to support the development of teaching materials. They used AI to prepare lesson materials, assessment questions, presentations, classroom activities, and to organize instructional content. Teachers used AI selectively, mainly for tasks they considered useful and relevant to their teaching needs. These findings align with previous research in higher education, showing that teachers use AI to prepare teaching materials, generate instructional content, and support lesson planning and other teaching-related activities (Zaim et al., 2024; Baggia et al., 2026).

The present findings extend this research by showing that teachers also used AI as a source of ideas when planning lessons and developing teaching materials, particularly when they needed alternative ideas or perspectives. Teachers explained that they turned to AI when they felt stuck, needed new ideas, or wanted to explore different ways of approaching lessons and classroom activities. This finding is consistent with Oh (2025), who observed that AI tools can complement teachers' creativity during lesson planning and the development of instructional materials. Therefore, with AI support, teachers can explore additional possibilities during the planning process while selecting and adapting ideas that are appropriate to their teaching context.

In addition, the findings show that teachers used AI to ease their workload and save time. This finding also reflects what Baggia et al. (2026) reported, as university teachers viewed AI as useful for working more efficiently and for producing materials that could be adapted to their needs. However, this benefit was not consistent across all situations. When the generated content met teachers' requirements, they found that AI was useful for reducing their workload and saving time. When the content did not meet their needs, however, they had to spend additional time editing the content, refining their prompts, or generating the material again. This denotes that using AI did not always reduce the time required to prepare teaching materials. The extent to which AI reduced teachers' workload was closely tied to the quality and relevance of its outputs and to how clearly teachers communicated their requirements in prompts. Therefore, unclear prompts or unsuitable content could result in additional checking, editing, and refinement.

Although AI can be used to prepare teaching materials more efficiently, the generated content still requires teacher review before it can be used. This connects with Baggia et al. (2026), whose review shows the importance of maintaining the human aspect of teaching when using AI. Zaim et al. (2024) also found that university lecturers reported benefits from using AI for teaching preparation and efficiency, while their findings point to challenges in ensuring that AI-generated outputs align with educational standards and teaching practices. In the present study, teachers reviewed and adapted AI-generated material before incorporating it into their teaching, particularly when the content required correction or adjustment. The usefulness of AI in preparing teaching materials depended on teachers' ability to judge whether the generated content was suitable for their particular teaching context.

Teachers' experiences with errors in AI-generated content further explain why their involvement remained necessary. Teachers reported finding factual mistakes, incorrect references, spelling errors, and other problems in AI-generated responses. These experiences showed that AI-generated information could not always be used without verification. Teachers had to review the information and determine whether it was accurate enough to include in their teaching materials. This was particularly important because an error that is not detected by the teacher can become part of the material presented to students. The findings therefore show that teachers' role was not limited to using AI to produce content; their knowledge was also needed to assess the information before it was used in teaching. This finding is consistent with Han et al.'s (2025) concern about the need to critically evaluate AI-generated content in education. However, while Han et al. (2025) focus on evaluating students' work in relation to AI-generated

content, the present study shows that a similar evaluation is also needed when teachers use AI to develop their own teaching materials.

Subject knowledge was particularly important when teachers had to identify these errors**.** Teachers explained that their subject knowledge (content knowledge) helped them to recognize when AI-generated information was incorrect or questionable. AI may generate content that appears reasonable, but teachers still need sufficient subject knowledge to recognize when the information is inaccurate or does not fit the subject. In this study, teachers therefore drew on their subject knowledge to assess the quality of AI-generated information and decide whether it could be accepted, checked, or corrected. Supporting this, Georgiou et al. (2026) emphasize the importance of teachers drawing on technological, pedagogical, and content knowledge when working with generative AI. The present findings provide a more specific illustration of this process by showing how content knowledge supported teachers in checking and validating AI-generated information. This relationship can be understood through the TPACK framework (Mishra & Koehler, 2006), which explains how technological, pedagogical, and content knowledge interact in teachers' use of technology.

Teachers also relied on their professional judgment when deciding how AI-generated information could be incorporated into their teaching materials. Their decisions involved more than identifying factual errors; they also considered whether the content was appropriate for their particular teaching purpose and context. When the material did not meet their needs, teachers refined the content, adjusted their prompts, or made changes themselves. This connects with Mota et al. (2026), whose review highlights the continuing role of teachers in evaluating AI use and adapting their teaching practices. In the present study, professional judgment was evident in the decisions teachers made after reviewing AI-generated content. Teachers remained responsible for deciding not only whether the information was reliable, but also whether it was appropriate for their intended teaching purpose.

It is essential for teachers to develop AI literacy to use AI effectively when creating AI-assisted teaching materials. However, due to limited formal guidance, many teachers have taken the initiative to develop their AI literacy through hands-on experience, learning by using AI in their teaching practice. They learned by experimenting with AI, practicing its use, learning from mistakes, using online resources, and discussing their experiences with colleagues. This finding mirrors what Kılıçkaya and Kic-Drgas (2025) and Oh (2025) reported, that teachers developed their ability to use AI through exploration and continued use. The present study shows that repeated use gradually helped teachers to become more confident in using AI and refine their teaching materials. Moreover, the findings show that developing AI literacy involved more than learning how to operate AI tools. As teachers became more familiar with AI, they also developed a better understanding of its limitations and learned to work with its outputs more carefully. They improved their prompts, reviewed the responses they received, and adjusted their approach based on what they encountered in practice. This suggests that practical engagement helped teachers develop not only greater confidence in using AI, but also a more critical understanding of its capabilities and limitations. Their learning was therefore closely connected to the challenges they encountered when integrating AI into their teaching practice.

However, much of this learning took place without formal institutional support. Daniel et al. (2026) similarly highlight the limited availability of comprehensive institutional frameworks and coordinated approaches for supporting AI integration in higher education. In the present study, the lack of formal guidance meant that teachers often had to take responsibility for developing their own understanding of AI. Their willingness to learn independently helped them develop practical knowledge, but it also raises questions about relying mainly on individual effort. As AI becomes more widely used in higher education, teachers may need more consistent opportunities to develop the knowledge and confidence required to use it effectively.

The increasing availability of AI has raised questions for teachers about how learning activities and assessments should be designed. In this study, teachers were concerned that students could use AI to complete assignments without fully engaging with the learning process. Their concern was not simply about students' use of AI, but about whether students had actually engaged with and understood the course content. This led teachers to reconsider how teaching materials and assessment activities could encourage students to think, participate, and develop their own understanding of the course content. In response to these concerns, teachers described changing some of the tasks they used with students. They used questions that required students to think about the content and provide their own responses, including higher-order thinking questions. These approaches were intended to encourage students to engage with the subject rather than rely entirely on AI responses. This suggests that students' use of AI has become something teachers need to consider when designing learning activities. The concern was therefore about how the design of an activity could encourage students to engage in the learning process rather than simply produce an answer. This finding adds to previous research documenting teachers' use of AI across a range of teaching-related activities, including lesson planning, instructional design, content development, and classroom activities (Zhai, 2023; Kılıçkaya & Kic-Drgas, 2025; Oh, 2025; Baggia et al., 2026; Zaim et al., 2024). In the present study, teachers' experiences illustrate that students' use of AI also influenced how teachers thought about learning activities and assessment. Teachers were not only considering how AI could help them prepare materials; they were also thinking about how students would use those materials and whether the activities would support the intended learning. This is particularly relevant to assessment activities in which teachers want students to demonstrate their own understanding of the subject.

The teachers' use of AI also involved having a clear purpose. They described deciding what they wanted to achieve before using AI and providing specific instructions when generating teaching materials. Some teachers noted that AI could provide additional suggestions that were not part of their original request. Although these suggestions could sometimes be useful, they could also distract teachers from the task at hand and require additional time. Having a clear purpose helped teachers determine which suggestions were relevant to the teaching task and which were not. Mota et al. (2026) similarly emphasize the continuing importance of teachers' professional judgment in deciding how AI is used in teaching practice. The present findings extend this understanding by showing that such judgment was exercised throughout teachers' interactions with AI. Teachers had to define what they wanted from the tool, assess whether the responses were relevant to their teaching purpose, and adapt or reject outputs that did not meet

their requirements. Purposeful use of AI, therefore, involved an ongoing process of directing and evaluating its outputs in relation to particular teaching goals. This highlights the continuing importance of teachers' professional judgment when AI is used to support the development of teaching materials.

These experiences show that using AI in the development of teaching materials involves decisions about teaching and learning, rather than simply about whether to use the technology. Teachers considered what students needed to learn, how they could engage with the content, and how AI could support these aims. This reflects the connection between technology and pedagogy described in the TPACK framework (Mishra & Koehler, 2006). In this study, teachers' use of AI was guided by the learning goals and the particular teaching situation in which the materials and activities would be used. Their pedagogical knowledge influenced how they considered the role of AI in designing activities that encouraged student engagement, reflection, and active involvement with the content. The findings, therefore, complement the TPACK perspective by showing how pedagogical considerations guided teachers' decisions about when and how to use AI to support meaningful learning.

While teachers placed considerable emphasis on the pedagogical quality of AI-generated materials, environmental sustainability received much less attention in their decision-making. Most teachers were familiar with issues such as plagiarism, academic integrity, ethics, and students' use of AI, but environmental sustainability was rarely part of their discussions about AI use. Several teachers also acknowledged that they had hardly thought about the environmental implications of AI before taking part in this study. This suggests that, for the teachers in this study, environmental sustainability had not yet become a regular consideration in their use of AI in teaching. Similar concerns have been raised by Leal Filho et al. (2025), who highlight the need to give further attention to the environmental and broader sustainability implications of AI in education

At the same time, teachers differed in how they understood and responded to the environmental implications of AI. While most teachers recognized the importance of considering environmental sustainability when developing teaching materials with AI, many teachers also expressed a need for further knowledge about the relationship between AI use and environmental impacts. Some teachers nevertheless described practical choices they had already made, such as avoiding unnecessary AI interactions, reusing existing teaching materials, limiting repeated prompts, and using AI when it served a clear educational purpose. These accounts suggest that environmental considerations were beginning to influence some teachers' decisions, although they had not yet become an established part of their teaching practice. The findings point to a gap between recognizing the importance of environmental sustainability and having sufficient knowledge to apply it when using AI for developing teaching materials.

This finding can be considered in relation to Clark's (2024) concept of Critical Climate Awareness, which emphasises the connection between everyday actions and broader environmental consequences. Some teachers in the present study were beginning to make this connection by considering whether an interaction with AI was necessary, whether existing materials could be reused, and whether repeated prompts or image generation could be avoided.

However, these choices were largely based on personal judgment and experience. Many teachers remained uncertain about the actual environmental effects of AI use, making it difficult for them to determine whether the practices they considered environmentally responsible would, in fact, reduce environmental impact. The findings, therefore, suggest that awareness of the issue alone may not be sufficient; teachers also need knowledge to connect their everyday choices about AI use with their broader environmental consequences.

Aczel et al. (2026) similarly emphasize that responsible AI use involves considering environmental consequences at both individual and institutional levels. This distinction is relevant to the present findings. At the individual level, some teachers were already making choices about when and how they used AI, such as avoiding unnecessary interactions and reusing existing materials. At the institutional level, however, teachers described a lack of discussion, guidance, and opportunities to learn about the environmental implications of AI. The findings, therefore, suggest that environmentally responsible use cannot depend entirely on individual teachers making their own judgments. Teachers need a clearer basis for understanding the environmental implications of different forms of AI use, while institutions also have a role in creating the conditions that support such understanding.

Teachers expressed a need for workshops, guidance, professional development, and opportunities to discuss the issue with colleagues. Their responses were not simply about learning how to use AI. They also wanted to understand the environmental implications of a technology they were already using in their teaching. This connects with Georgiou et al. (2026) and Daniel et al. (2026), who highlight the importance of institutional support and professional development in helping teachers develop their understanding of AI and make decisions about its use in higher education. In the present study, however, teachers described few opportunities to discuss the environmental implications of AI in their formal or informal professional settings. As a result, many were developing their understanding through their own experiences while still having limited knowledge of the environmental consequences of different forms of AI use. This helps explain why environmental considerations were present in some teachers' practices but had not yet become a consistent part of their decisions about using AI.

Some teachers also suggested that students should learn about the environmental implications of AI and that the topic could be included in education. This is consistent with Leal Filho et al. (2025), who emphasise the importance of considering sustainability in AI-supported teaching and learning. At the same time, teachers' own uncertainty is important in understanding this finding. Environmental sustainability was still relatively new in their discussions about AI, and their views about what students should learn were developing alongside their own understanding of the issue. Their responses suggest that environmental sustainability was beginning to be seen as relevant not only to how AI is used, but also to what teachers and students should understand about its wider consequences.

Strengthening greater awareness of these consequences could also help teachers make more deliberate choices about their use of AI. With a clearer understanding of the environmental effects associated with different forms of AI use, teachers may be better able to consider whether a particular use is necessary for a teaching purpose, whether existing materials can be reused, or whether some interactions with AI can be avoided. This would give greater meaning

to the practical choices some teachers were already making, such as reusing materials and limiting unnecessary interactions. In this sense, the issue is not simply whether teachers use AI, but whether they understand the consequences of their choices about how to use it.

A key contribution of this study is that it brings together teachers' pedagogical use of AI and their consideration of its environmental consequences. Previous research has often examined these issues separately, focusing either on how AI can support teaching and learning or on its environmental implications at technological and institutional levels. The present findings bring these perspectives closer to teachers' everyday practice. They show that teachers were already making decisions about how AI should be used in the development of teaching materials, while consideration of its environmental consequences was still emerging. Some teachers had begun to make practical choices based on their own understanding, but limited knowledge and discussion meant that these considerations were not yet consistent. The findings, therefore, suggest that responsible AI use in higher education needs to be considered not only in terms of how AI supports teaching, but also in relation to the wider environmental consequences of its use.

6.1 Limitations of the study

Several limitations should be considered when interpreting the findings of this study. One limitation of this study is that it was based on a relatively small qualitative sample of higher education teachers from Sweden, Denmark, and India. Although including teachers from different national contexts provided a range of perspectives and experiences, the findings cannot be generalized to all higher education settings or teacher populations. Instead, they provide an in-depth understanding of teachers' experiences within the contexts represented in this study.

The study also relied on teachers' self-reported experiences and reflections, with findings based on participants' accounts of AI-supported teaching practices and not on direct observations of classroom practices. Consequently, the findings reflect how teachers perceived and described their use of AI, which may differ from their actual classroom practices. In addition, teachers may have interpreted AI use, responsible practices, and environmental sustainability differently depending on their individual experiences and levels of awareness.

In addition, the study focused on higher education teachers and did not include the perspectives of other stakeholders, such as students, institutional leaders, policymakers, or AI developers. Including these groups could have provided a more comprehensive understanding of AI integration, institutional decision-making, and environmental sustainability in higher education.

Another limitation is that the study did not examine the environmental impacts of specific AI models or systems, such as differences in energy consumption, carbon emissions, or resource use. Instead, it explored teachers' awareness and perceptions of environmental sustainability in relation to AI-supported teaching material development. Therefore, the findings should be

interpreted as reflecting teachers' perceptions of environmental impacts and should not be considered objective assessments of the environmental impacts associated with particular AI technologies.

In addition, all interviews were conducted in English, which was not the first language of either the researcher or many of the teachers. Although participants were able to communicate their experiences, language differences may have influenced how some ideas were expressed, meaning that certain nuances or deeper reflections may not have been fully captured.

Environmental sustainability and AI use were also relatively unknown topics for many teachers. Several participants explained that they had rarely reflected on the environmental implications of AI before participating in the interviews. As a result, some responses appeared exploratory as teachers considered these issues during the discussion. While this provided valuable insights into emerging sustainability awareness, it may also have limited the depth of reflection on the environmental implications of AI use.

As with all qualitative research, the interpretation of interview data involved the researcher's analytical judgment during coding and theme development. Although the analysis was grounded in teachers' accounts and developed through repeated engagement with the data, alternative interpretations of some responses were also possible.

Finally, AI technologies and their use in education continue to evolve rapidly. Teachers' knowledge, institutional policies, and available AI tools are likely to change over time. Consequently, the findings reflect teachers' experiences at the time the study was conducted and may differ as AI continues to develop.

7. Conclusion

This study examined how higher education teachers use AI in developing teaching materials and how they understand the environmental sustainability of this use. The findings show that AI has become a practical support in teachers' preparation of teaching materials, but its use involves more than simply generating content. Teachers decided when AI was useful, checked what it produced, and made changes based on their subject knowledge, teaching goals, and understanding of their students. The usefulness of AI in teaching material development, therefore, depended not only on what the technology could produce but also on how teachers evaluated and adapted it.

The findings also show that teachers developed their ability to use AI mainly through experience. They learned by trying different approaches, learning from mistakes, adjusting prompts, checking responses, and discussing their experiences with colleagues. Through this process, they became more familiar with what AI could do and where its limitations lay. However, formal guidance was limited, which meant that much of this learning depended on teachers' own initiative. This points to a gap between the growing use of AI in teaching and

the opportunities available to teachers to develop their knowledge and skills in a more structured way.

Another important finding concerns the place of environmental sustainability in teachers' use of AI. Teachers were generally more familiar with issues such as academic integrity, plagiarism, ethics, and student use of AI, while the environmental effects of AI received much less attention. Some teachers had already begun to make practical choices, such as avoiding unnecessary use, reusing existing materials, and limiting repeated interactions with AI. However, these choices were mostly based on personal understanding, and many teachers remained unsure about the actual environmental consequences of different forms of AI use. Environmental sustainability was therefore recognized as important, but it had not yet become a regular part of teachers' decisions about using AI.

The findings also bring these aspects together. Teachers were already making careful decisions about the accuracy, relevance, and pedagogical value of AI-generated materials, while environmental considerations were still developing. This difference is important because it shows that responsible use of AI in teaching cannot be considered only in terms of whether AI produces useful content or supports teaching efficiently. It also involves understanding the wider consequences of using the technology and considering whether its use is necessary for a particular teaching purpose.

The study concludes that teachers remain central to the use of AI in the development of teaching materials. AI can support their work, but teachers determine how the technology is used, how its outputs are evaluated, and whether the resulting materials are appropriate for their teaching. At the same time, the findings point to the need for greater attention to the environmental side of AI use. Developing teachers' AI literacy alongside their understanding of environmental implications can provide a stronger basis for decisions about when and how AI should be used in teaching. This study consequently adds to the discussion of responsible AI in higher education by showing that pedagogical judgment and environmental awareness need to be considered together when AI becomes part of everyday teaching practice.

7.1 Societal Relevance

This study addresses a timely issue as artificial intelligence (AI) becomes increasingly integrated into higher education. As the use of AI expands, its implications extend beyond teaching efficiency and include questions about how teachers use the technology, how they make decisions about AI-generated content, and how they consider its wider consequences. The findings provide insight into these issues through teachers' experiences and point to areas where further support may be needed, particularly in AI literacy, pedagogical decision-making, and environmental sustainability.

The findings are relevant to higher education institutions because they point to the need for support that addresses the different aspects of teachers' use of AI. This includes opportunities to develop AI literacy, support for pedagogical decision-making, and greater awareness of the

environmental implications of AI use. In this study, teachers were already taking responsibility for checking and adapting AI-generated materials, while their understanding of the environmental implications of AI use was more limited. Institutional support could provide opportunities for teachers to develop knowledge in this less familiar area and consider environmental issues alongside the pedagogical purposes of AI use. This is consistent with broader discussions of responsible AI, which emphasize the need to consider the consequences of AI use alongside its practical benefits (Aczel et al., 2026).

The findings are also relevant to teachers' everyday practice. Teachers' experiences show that using AI effectively involves keeping the teaching purpose in view, evaluating AI-generated content, and making changes when the material does not meet the needs of the teaching situation. Environmental considerations add another aspect to these decisions. Some teachers had already begun to avoid unnecessary use, reuse existing materials, and limit repeated interactions with AI, although these choices were often based on their own understanding of the issue. Greater knowledge of the environmental consequences of AI could help teachers consider such choices more deliberately alongside their pedagogical needs. This is consistent with Leal Filho et al. (2025), who emphasize the need to give greater attention to the sustainability implications of AI as its use expands in education.

The study also has relevance for students and the wider educational community. Some teachers expressed a need for greater awareness of the environmental implications of AI among both educators and students, and suggested that the topic could receive greater attention in education. This points to an opportunity to bring environmental considerations into discussions of AI beyond questions of academic integrity, ethics, and educational effectiveness. Greater awareness among teachers and students could support broader discussion of the environmental consequences of digital technologies within higher education.

More broadly, this study contributes to discussions about responsible AI use in higher education by bringing pedagogical and environmental considerations together. The findings show that teachers are already making decisions about the accuracy, relevance, and educational value of AI-generated materials, while environmental sustainability remains less established in these decisions. The societal relevance of this finding lies in recognizing that the growing use of AI in education has consequences that extend beyond the immediate teaching task. Considering these consequences alongside educational needs can support a more balanced approach to AI use, in which the benefits of the technology are considered together with its wider environmental implications.

7.2 Implications

This study has several practical implications for higher education teachers and institutions regarding the responsible use of AI in teaching material development. These implications should, however, be considered in relation to the institutional, disciplinary, and teaching contexts in which AI is used.

For teachers, the findings highlight the importance of maintaining professional judgment throughout the development of AI-supported teaching materials. AI-generated content needs to

be reviewed, adapted, and considered in relation to learning objectives, students' needs, and disciplinary requirements. This is consistent with McGrath et al. (2023), who emphasize the continuing responsibility of university teachers when making decisions about the use of AI in higher education. The findings also suggest that teachers can benefit from using AI with a clear teaching purpose. Defining what is needed before using AI and considering how the resulting material will support student learning can help keep its use connected to the teaching purpose. Ongoing development of AI literacy is also important, particularly in evaluating AI-generated content, recognizing its limitations, and deciding when and how it can appropriately support teaching (Georgiou et al., 2026).

For higher education institutions and educational leaders, the findings point to the need for professional development that reflects how teachers are actually learning to use AI. Since many participants developed their knowledge through experimentation, self-directed learning, and discussions with colleagues, institutions could provide more structured opportunities for teachers to learn from one another and develop their practice. Professional development could include practical activities focused on evaluating AI-generated content, developing appropriate prompts, adapting AI-supported materials to different teaching contexts, and considering the pedagogical issues that arise when students also use AI.

The findings also point to a need to include environmental sustainability within institutional discussions of AI. Many teachers had limited knowledge of the environmental implications of AI, even though they recognized that these issues were important. Institutions could therefore include environmental considerations in AI-related professional development, guidance, and digital literacy initiatives. Such support could help teachers develop a clearer understanding of the environmental implications of different forms of AI use and consider these alongside their teaching needs. This is consistent with broader discussions of responsible AI that emphasize attention to environmental consequences as part of responsible use (Aczel et al., 2026).

The findings further suggest that institutional guidance should address more than the technical use of AI. Teachers need support in deciding when AI is useful, how to evaluate its outputs, and how to keep its use connected to the purpose of teaching. At the same time, environmental considerations need to become more visible within these discussions. Providing opportunities for teachers to develop this knowledge and share their experiences may help institutions develop guidance that is more closely connected to everyday teaching practice.

Overall, the practical implication is not that teachers should use AI more or less, but that they need a stronger basis for deciding when and how its use is appropriate. This involves combining teachers' existing professional judgment with developing AI literacy and a better understanding of the environmental implications of AI. Such an approach can support more considered use of AI in teaching material development while leaving room for differences across disciplines, teaching contexts, and institutional settings.

7.3 Future Research Directions

This study provides insight into how higher education teachers use AI in teaching material development and how they perceive environmental sustainability in relation to these practices. As a qualitative study, the findings reflect teachers' experiences within the educational contexts included in the study and at a time when AI technologies are continuing to develop. Teachers' practices, AI literacy, and understanding of environmental sustainability may therefore change as AI becomes more established in higher education. Further research can build on these findings by examining how these aspects develop across different contexts and over time.

Research across a wider range of educational contexts would be valuable, including different academic disciplines, types of higher education institutions, countries, and educational settings. Such research could examine whether similar patterns emerge in teachers' use of AI, development of AI literacy, and consideration of environmental sustainability. Comparative studies could also provide greater insight into how institutional, disciplinary, and cultural contexts influence teachers' decisions about AI use.

Since this study focused on higher education teachers, future research could also include the perspectives of students, educational leaders, policymakers, and other relevant stakeholders. Including these perspectives would provide a broader understanding of how AI affects teaching and learning and how decisions about its use are made within higher education. In particular, research involving students could examine their experiences with AI-supported teaching materials and explore how such materials relate to learning, engagement, and critical thinking.

Further research could also follow teachers over time to examine how their AI literacy, pedagogical practices, and understanding of environmental sustainability develop as their experience with AI increases. Longitudinal research could provide insight into whether the practices identified in this study become more established or change as teachers gain experience and as AI technologies develop. Combining qualitative interviews with classroom observations or analysis of AI-supported teaching materials could also provide a closer connection between teachers' reported experiences and their actual teaching practices.

The findings also raise questions about the role of institutional support. Future research could examine how different forms of professional development, institutional guidance, and AI policies influence teachers' use of AI. In particular, studies could explore whether professional development that includes both pedagogical and environmental considerations helps teachers make more informed decisions about when and how AI is used in teaching material development. This could provide evidence about the types of institutional support that are most useful in practice.

Finally, this study examined teachers' perceptions of environmental sustainability rather than measuring the actual environmental impacts of their AI use. Future research could therefore connect teachers' experiences with measures of environmental impact, such as energy consumption, carbon emissions, water use, or other relevant indicators associated with AI-supported activities. This would help connect teachers' perceptions and everyday practices with the environmental consequences of AI use and provide a more complete understanding of how

AI can be used in higher education while considering both educational needs and environmental sustainability.

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Appendix

Appendix A

Section 1. Background and Teaching Context

1. Can you describe your current teaching role and the subjects or courses you teach?
2. How would you describe your general experience with digital technologies in teaching and learning?

Section 2. AI Use in Teaching Material Development

3. What types of AI tools do you currently use when creating teaching materials?
4. Can you describe a recent example of how you used AI to develop teaching materials or learning activities?
5. How does the use of AI influence the time you spend preparing teaching materials?
6. In what ways do AI tools influence your process of designing teaching materials and learning activities?
7. How do you decide when it is appropriate to use AI in teaching materials?
8. Do you follow any personal or institutional strategies, guidelines, or best practices when using AI?
9. In what ways do AI-generated materials support or challenge your teaching goals?
10. Do you usually modify or refine AI-generated content before using it in teaching? If yes, how?
11. How do students influence your decisions when using AI to create teaching materials?

Section 3. AI Literacy, Responsibility, and Educational Challenges

12. How confident do you feel in your ability to use AI tools effectively in teaching?

13. Have you experienced any challenges or concerns related to AI use in education, such as accuracy, bias, plagiarism, or student dependency?
14. Do you feel that you have sufficient knowledge or institutional support to use AI responsibly in teaching?
15. What kinds of support, training, or guidance would help teachers use AI more effectively?

Section 4. Sustainability and Environmental Awareness

16. When you think about environmental sustainability in education, what does it mean to you?
17. Have you ever reflected on the environmental impact of digital technologies used in teaching?
18. Have you considered the environmental impact of AI technologies specifically?
19. Have you reflected on issues such as energy use or digital resource consumption when using AI tools?
20. Do you take any actions or make any choices to reduce the environmental impact of your AI-supported teaching practices?
21. To what extent do you feel responsible for the environmental impact of AI technologies used in teaching?
22. Do you experience any tensions between the pedagogical benefits of AI and environmental sustainability considerations?
23. What kinds of support or guidance would help teachers use AI more sustainably?
24. How important do you think environmental sustainability should be in future discussions about AI in education?

Section 5. Future Perspectives and Final Reflections

25. How do you see the role of AI in teaching developing in the future?
26. Is there anything else you would like to add regarding AI use, teaching material development, or environmental sustainability?

Appendix B

Table 2

Example of Initial Codes, Grouped Codes, and Themes from Interview Data

Interview Extract	Initial Code	Grouped Code	Final Theme
"I use AI to assist me in creating the class test questionnaires, and made the preparation process easier and more efficient." (T5)	Assessment preparation	Creating teaching materials	AI as a Pedagogical Support Tool
"I use it to create cases (realistic scenarios) for my teaching." (T1)	Case generation	Creating teaching materials	AI as a Pedagogical Support Tool
"I recently used AI to create a PowerPoint for the Romantic Age of English literature." (T11)	Presentation development	Creating teaching materials	AI as a Pedagogical Support Tool
"I mostly turn to AI when I'm low on motivation, running out of ideas, and feeling stuck." (T1)	Brainstorming ideas	Generating ideas and lesson planning	AI as a Pedagogical Support Tool
"I could explore more from many perspectives to include activities in my class." (T7)	Exploring perspectives	Generating ideas and lesson planning	AI as a Pedagogical Support Tool
"It saves time, especially when you have a lot to do." (T1)	Time saving	Saving time and managing workload	AI as a Pedagogical Support Tool
"I always try to read and check the provided materials, and if needed, I recreate and modify my prompt." (T2)	Reviewing AI-generated content	Reviewing and refining AI-generated content	Teacher Expertise in Evaluating and Validating AI-Generated Content

"I always check the generated content, and I read it twice very slowly." (T5)	Verifying AI-generated content	Reviewing and refining AI-generated content	Teacher Expertise in Evaluating and Validating AI-Generated Content
"When I saw that spelled it wrong, I needed to go through the whole text." (T1)	Identifying AI errors	Recognising errors and reliability concerns	Teacher Expertise in Evaluating and Validating AI-Generated Content
"We should be very conscious of the content we generate as it can make mistakes." (T7)	Reliability concerns	Recognising errors and reliability concerns	Teacher Expertise in Evaluating and Validating AI-Generated Content
"You should have the knowledge within the subject to identify the mistakes." (T1)	Subject knowledge	Subject knowledge and professional judgement	Teacher Expertise in Evaluating and Validating AI-Generated Content
"We can't use AI blindly..." (T9)	Critical evaluation	Subject knowledge and professional judgement	Teacher Expertise in Evaluating and Validating AI-Generated Content
"I learn from the mistake and try not to make it next time." (T2)	Learning through mistakes	Learning through experimentation and practice	Developing AI Literacy Through Practice, Adaptation, and Support
"From mistakes, I learned how to give detailed and refined prompts." (T8)	Prompt refinement	Learning through experimentation and practice	Developing AI Literacy Through Practice, Adaptation, and Support
"I feel like I am forced to use AI because you have to know how it works." (T3)	Professional adaptation	Adapting to a changing educational environment	Developing AI Literacy Through Practice, Adaptation, and Support
"AI is a growing term... otherwise I feel I am far away from the current trend." (T4)	Keeping up with educational change	Adapting to a changing educational environment	Developing AI Literacy Through Practice, Adaptation, and Support

"We need more knowledge on this topic." (T5)	Need for AI knowledge	Need for training, guidance, and institutional support	Developing AI Literacy Through Practice, Adaptation, and Support
"A teachers' forum would be beneficial..." (T7)	Professional development	Need for training, guidance, and institutional support	Developing AI Literacy Through Practice, Adaptation, and Support
"Students use AI for their assignments, so it is difficult to determine whether they have used it." (T9)	Student reliance on AI	Designing AI-supported learning activities for meaningful learning	Pedagogical Considerations in AI-Supported Teaching Material Development
"The ultimate aim for the assignments is student learning… but now there is doubt whether learning is really happening." (T12)	Maintaining meaningful assessment	Designing AI-supported learning activities for meaningful learning	Pedagogical Considerations in AI-Supported Teaching Material Development
"I make questionnaires with higher-order thinking levels..." (T9)	Higher-order learning activities	Designing AI-supported learning activities for meaningful learning	Pedagogical Considerations in AI-Supported Teaching Material Development
"I use AI tools with defined goals." (T12)	Purposeful AI use	Purposeful pedagogical use of AI	Pedagogical Considerations in AI-Supported Teaching Material Development
"Use AI with a clear purpose." (T9)	Goal-oriented AI use	Purposeful pedagogical use of AI	Pedagogical Considerations in AI-Supported Teaching Material Development
"I was wondering what the connection between the environment and AI is." (T10)	Limited environmental awareness	Limited awareness of environmental impact	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching

			Material Development
"I really don't know how it is connected with the environmental consequences." (T11)	Limited environmental knowledge	Limited awareness of environmental impact	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development
"I don't use it for fun or unnecessary things. I always use it with a clear purpose." (T8)	Purposeful AI use for environmental consideration	Environmental considerations in teachers' AI use	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development
"I don't generate new AI images every time. I reuse existing ones." (T3)	Reusing AI resources	Environmental considerations in teachers' AI use	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development
"We should include AI and its impact awareness topics in the curriculum." (T10)	Sustainability education	Need for sustainability education and institutional awareness	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development
"We need workshops, training, or guidance from the institution side..." (T11)	Institutional support for sustainability	Need for sustainability education and institutional awareness	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development

Initial Codes	**Grouped Code**	**Final Theme**
Lesson planning, presentations, assessment	Creating teaching materials; Generating ideas and lesson planning; Managing workload	AI as a Pedagogical Support Tool

preparation, brainstorming, workload reduction		
Reviewing AI content, verifying outputs, recognising errors, subject knowledge, critical evaluation	Reviewing and refining AI-generated content; Recognising errors and reliability concerns; Professional judgement	Teacher Expertise in Evaluating and Validating AI-Generated Content
Learning through mistakes, experimentation, adapting to AI, institutional support	Learning through experimentation; Adapting to a changing educational environment; Institutional support	Developing AI Literacy Through Practice, Adaptation, and Support
Student engagement, meaningful learning, purposeful AI use	Designing meaningful learning activities; Purposeful pedagogical use of AI	Pedagogical Considerations in AI-Supported Teaching Material Development
Limited environmental awareness, conscious AI use, sustainability education	Limited awareness of environmental impact; Environmental considerations in AI use; Need for sustainability education and institutional awareness	Teachers' Perceptions of Environmental Sustainability in AI-Supported Teaching Material Development

Appendix C

Participant Information and Informed Consent Letter

Title of the Study

Teachers' Use of Artificial Intelligence in the Creation of Teaching Materials and Consideration of Environmental Sustainability in Education.

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Master Program in Digital Learning

Halmstad University athiraambilijeeva@gmail.com

Invitation to Participate

You are invited to take part in a research study that explores how teachers in higher education use artificial intelligence (AI) tools when creating teaching materials, and how environmental sustainability is understood, considered, or addressed within these practices. You are being invited because of your experience using AI tools in teaching or teaching material design.

Before deciding whether to participate, please read the information below carefully. You are welcome to ask questions at any time.

Purpose of the Study

The purpose of this study is to gain a deeper understanding of:

- How teachers incorporate AI tools into the creation of teaching materials and learning activities, and
- How environmental sustainability is reflected upon, considered, or overlooked in these practices.

The study aims to contribute knowledge that can support more responsible and environmentally conscious use of AI in education.

What Participation Involves

If you agree to participate:

- You will take part in a semi-structured interview lasting approximately 45–60 minutes.
- The interview will be conducted online or in person, depending on your preference.
- The interview will focus on your experiences, reflections, and decision-making related to AI use in teaching and sustainability considerations.
- With your consent, the interview will be audio-recorded to ensure accurate transcription.

There are no right or wrong answers. The study seeks your personal experiences and perspectives.

Voluntary Participation and Right to Withdraw

Participation in this study is entirely voluntary.

- You may decline to answer any question.
- You may withdraw from the study at any time, without providing a reason and without any negative consequences.
- If you choose to withdraw, all data related to you will be deleted and will not be used in the study.

Confidentiality and Data Management

- All information you provide will be treated confidentially.
- Your identity will be anonymised in transcripts, analyses, and publications.
- No names or identifying details will appear in the thesis or any related publications.
- Audio recordings and transcripts will be stored securely in password-protected files and accessed only by the researcher.
- Data will be handled in accordance with ethical research guidelines and data protection regulations.

Potential Benefits and Risks

There are no anticipated risks associated with participating in this study. Some questions may encourage reflection on teaching practices or sustainability issues; you are free to skip any question you prefer not to answer.

While there may be no direct personal benefit, your participation will contribute to research on AI use, teaching practices, and sustainability in higher education.

Ethical Approval

This study follows the principles of good research practice and ethical guidelines for educational research. Ethical responsibility will be maintained throughout the research process.

The study is part of my master's program in Digital Learning at the School of Learning, Humanities, and Society at Halmstad University. The responsible teacher is Emma Edstrand (emma.edstrand@hh.se). Your personal data will be processed within the study only with your consent and in accordance with the GDPR. Your participation in the study is entirely voluntary, and you can revoke your consent at any time, without stating a reason. Withdrawing your

consent will not affect processing that occurred before your revocation. You may also ask for a copy of your personal data at any time. The recordings and transcriptions will be processed within the Microsoft Office 365 cloud service. The university has a data processor agreement with Microsoft. Unauthorized individuals will not have access to your personal data. Your personal data will be deleted when my examination paper is approved.

Consent

I confirm that:

- You have read and understood the information provided above.
- You have had the opportunity to ask questions.
- You voluntarily agree to participate in this study.
- You consent to the audio recording of the interview for research purposes.

Participant Name:

Signature:

Date: